

01

Coding with AI



AI-Driven Chatbots

- Chatbots powered by AI use large language models (LLMs)
- They understand and generate human-like text
- Interaction happens through natural language dialog



From Conversation to Learning

- Share ideas, tell stories, or role-play scenarios
- Provide quick explanations of concepts

Tell me a story about robots.



Explain quantum entanglement in simple terms.



Help me learn basic Python loops.

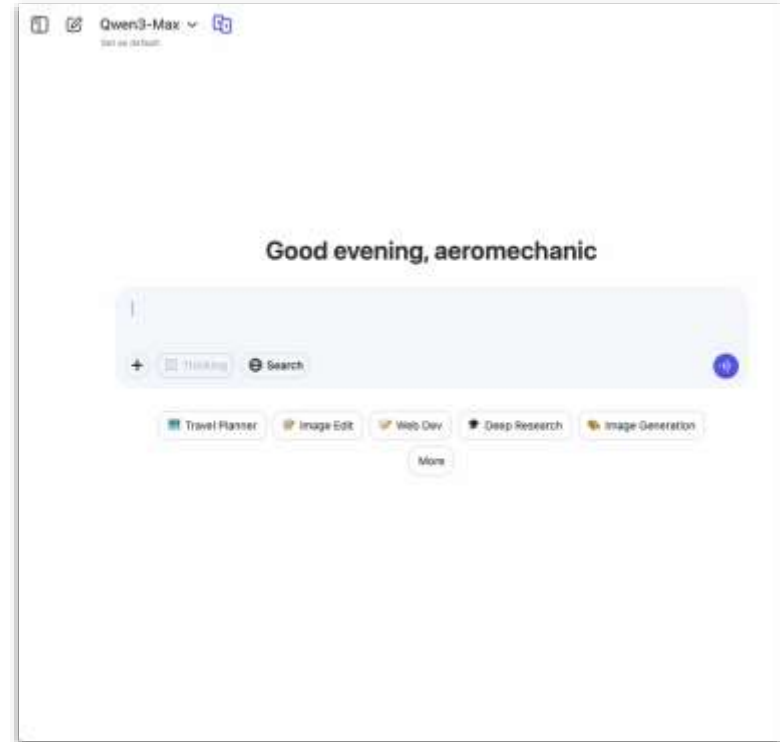


Coding Help

- Write code snippets in many programming languages
- Debug errors and suggest fixes
- Explain programming concepts with examples

Write a Python function to reverse a string.

- **Input:** abcde
- **Output:** edcba



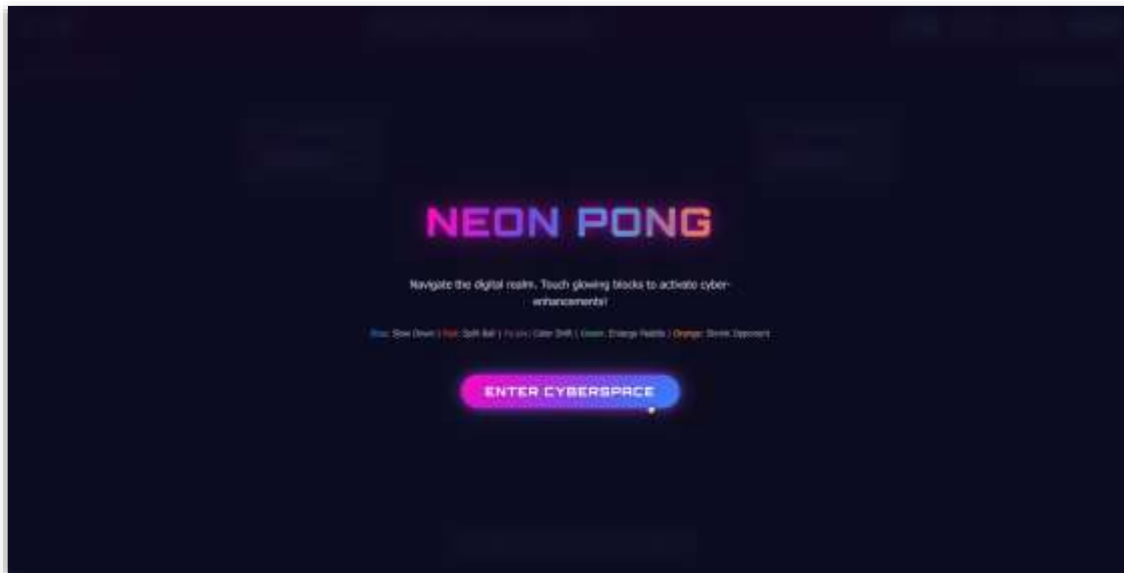
Coding Help – Hands on

1. Open <https://chat.qwen.ai> and login with your account.
2. Type in “**Write a Python function to reverse a string**”.
3. Copy & paste the *reverse_string* function to the vscode and run a test.



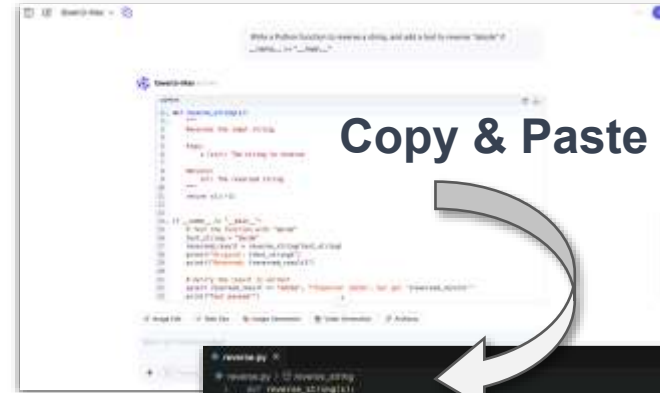
Production-Level Code

- Generate structured, reusable modules
- Provide documentation and comments
- Suggest design patterns and best practices

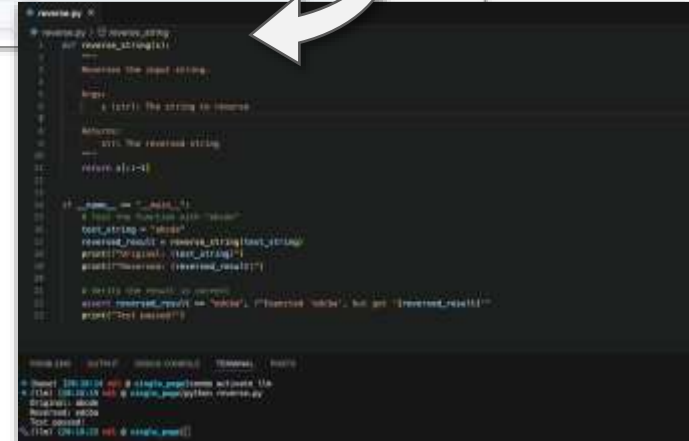


Coding with Chatbot

1. Open <https://chat.gwen.ai/> and login with your account.
2. Type in **the coding requirement**.
3. Copy & paste the result to a file and run it.
4. Found error?



Copy & Paste

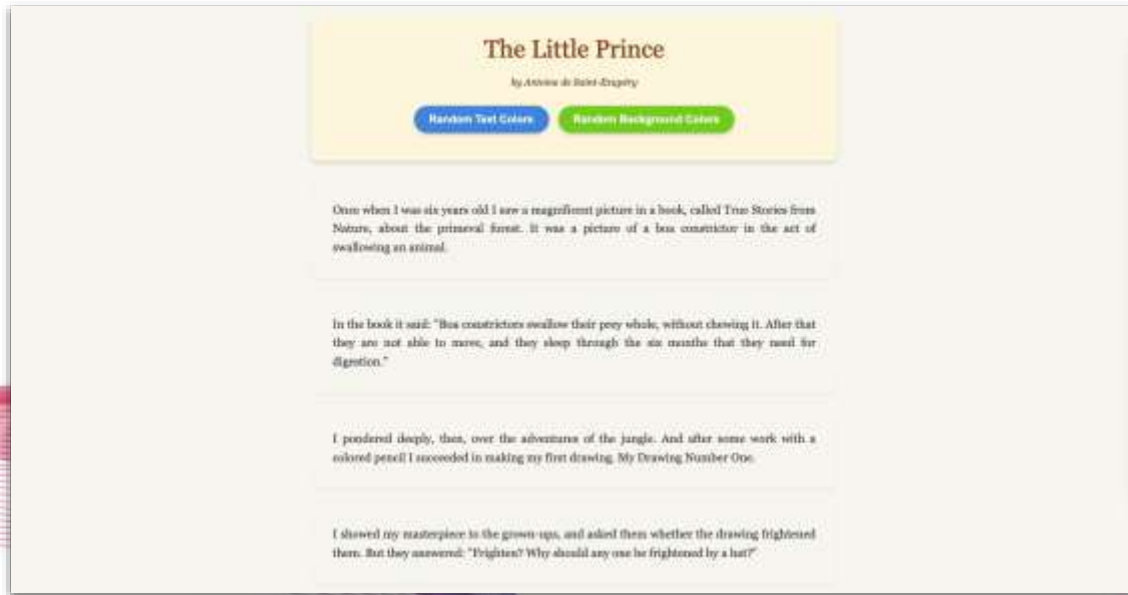


The screenshot shows a terminal window with the same Python code pasted into it. The code is executed, and the output is displayed at the bottom.

```
reverse.py 1 reverse_string  
1 def reverse_string(s):  
2     """Reverse the input string.  
3     Args:  
4         s (str): The string to reverse.  
5     Returns:  
6         str: The reversed string.  
7     """  
8     return s[::-1]  
9  
10 if __name__ == '__main__':  
11     # Test the function with 'abcde'  
12     test_string = 'abcde'  
13     reversed_result = reverse_string(test_string)  
14     print(f'Original: {test_string}')  
15     print(f'Reversed: {reversed_result}')  
16  
17 # Verify the result by comparing  
18 assert reversed_result == 'edcba', f'Expected 'edcba', but got '{reversed_result}'  
19 print('Test passed!')
```

```
reverse.py 1 reverse_string  
1 def reverse_string(s):  
2     """Reverse the input string.  
3     Args:  
4         s (str): The string to reverse.  
5     Returns:  
6         str: The reversed string.  
7     """  
8     return s[::-1]  
9  
10 if __name__ == '__main__':  
11     # Test the function with 'abcde'  
12     test_string = 'abcde'  
13     reversed_result = reverse_string(test_string)  
14     print(f'Original: {test_string}')  
15     print(f'Reversed: {reversed_result}')  
16  
17 # Verify the result by comparing  
18 assert reversed_result == 'edcba', f'Expected 'edcba', but got '{reversed_result}'  
19 print('Test passed!')
```

Build A Colorful Webpage



Introduction to HTML

- HTML: **HyperText Markup Language**
- The backbone of all web pages
- Defines the structure and content of a page
- Works together with
 - CSS for **styling**, and
 - JavaScript for **behavior control**

- Create a folder named “VibeCoding” in your home directory.
- In VSCode, click “Open Folder” under “File” menu, navigate to the created directory “VibeCoding” and click “select”.
- Select “New Folder” button, and create a new folder with name “lesson01”
- Select “New File” in “File” menu, and create a new file with name “hello.html”.
- Copy the html code into “hello.html”.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hello</title>
</head>
<body>
  Hello, world!
</body>
</html>
```

Open in Browser



Hello, world!

The Building Blocks

- HTML uses **elements** (tags) to describe content
- Elements have an **opening tag** and a **closing tag**
- Nesting allows creating structured layouts

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hello</title>
</head>
<body>
  <h1>Hello, world!</h1>
  <p>This is a paragraph of text.</p>
  <a href="https://www.example.com">This is a link.</a>
</body>
</html>
```

Hello, world!

This is a paragraph of text.

[This is a link.](#)

Common Elements

- Headings: `<h1> ... <h6>`
- Paragraph: `<p>`
- Image: ``
- Link: `<a href="https://example.com">Visit</a>`
- Input field: `<input type="text">`
- Button: `<button>`
- **Division: `<div>`, a generic container**

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hello</title>
</head>
<body>
  <h1>Hello, world!</h1>
  <div>
    <p>This is a paragraph of text.</p>
    <a href="https://www.example.com">This is a link.</a>
  </div>
</body>
</html>
```

Adding Style with CSS

- CSS: Cascading Style Sheets
- Controls **look and feel**
- Add styles inline, in `<style>` tags, or as external files

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hello</title>
</head>
<body>
  <h1>Hello, world!</h1>
  <p>This is a paragraph before the div.</p>
  <div style="color:red">
    <p>This is a paragraph inside the div.</p>
  </div>
  <p>This is a paragraph after the div.</p>
</body>
</html>
```

Hello, world!

This is a paragraph before the div.

This is a paragraph inside the div.

This is a paragraph after the div.

Adding Style with CSS

- CSS: Cascading Style Sheets
- Controls **look and feel**
- Add styles inline, in `<style>` tags, or as external files

Hello, world!

This is a paragraph before the div.

This is a paragraph inside the div.

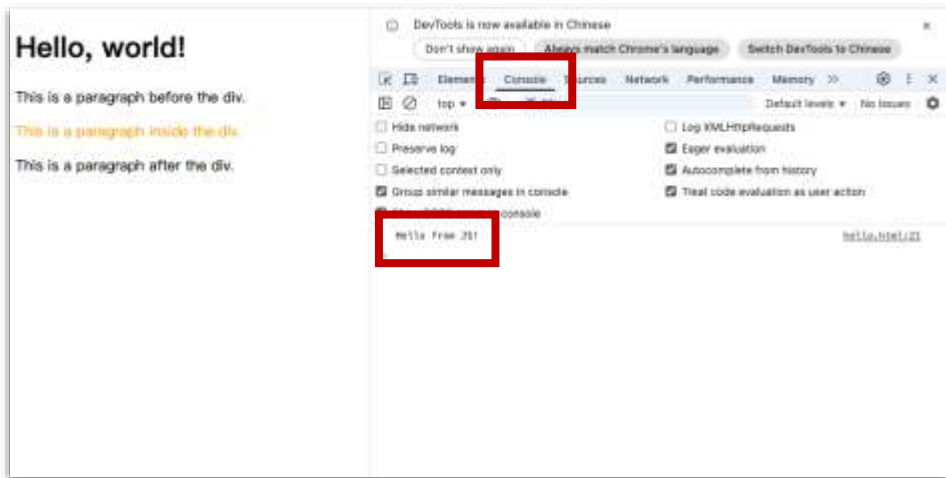
This is a paragraph after the div.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hello</title>
  <style>
    .mydiv {
      color: orange;
    }
  </style>
</head>
<body>
  <h1>Hello, world!</h1>
  <p>This is a paragraph before the div.</p>
  <div class="mydiv">
    <p>This is a paragraph inside the div.</p>
  </div>
  <p>This is a paragraph after the div.</p>
</body>
</html>
```

JavaScript in HTML

- JavaScript controls behavior
- Add with `<script>` tag

In browser, use F12 to open the developer tools



```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hello</title>
  <style>
    .mydiv {
      color: orange;
    }
  </style>
</head>
<body>
  <h1>Hello, world!</h1>
  <p>This is a paragraph before the div.</p>
  <div class="mydiv">
    <p>This is a paragraph inside the div.</p>
  </div>
  <p>This is a paragraph after the div.</p>
  <script>
    console.log(" Hello from JS!");
  </script>
</body>
</html>
```

JavaScript in HTML

- JavaScript controls behavior
- Add with `<script>` tag

Hello, world!

This is a paragraph before the div.

This is a paragraph inside the div.

This is a paragraph after the div.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
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  <style>
    .mydiv {
      color: orange;
    }
  </style>
</head>
<body>
  <h1>Hello, world!</h1>
  <p>This is a paragraph before the div.</p>
  <div class="mydiv">
    <p>This is a paragraph inside the div.</p>
  </div>
  <p>This is a paragraph after the div.</p>
  <script>
    console.log(" Hello from JS!");
    document.getElementsByClassName("mydiv")[0].style.color = "green";
  </script>
</body>
</html>
```

JavaScript in HTML

- JavaScript controls behavior
- Add with `<script>` tag

Hello, world!

This is a paragraph before the div.

This is a paragraph inside the div.

This is a paragraph after the div.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hello</title>
  <style>
    .mydiv {
      color: orange;
    }
  </style>
</head>
<body>
  <h1>Hello, world!</h1>
  <p>This is a paragraph before the div.</p>
  <div class="mydiv">
    <p>This is a paragraph inside the div.</p>
  </div>
  <p>This is a paragraph after the div.</p>
  <script>
    console.log(" Hello from JS!");
    document.getElementsByClassName("mydiv")[0].style.backgroundColor = "green";
  </script>
</body>
</html>
```

JavaScript in HTML

- JavaScript controls behavior
- Add with `<script>` tag

Hello, world!

This is a paragraph before the div.

This is a paragraph inside the div.

This is a paragraph after the div.

This is a paragraph inside the second div.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hello</title>
  <style>
    .mydiv {
      color: orange;
    }
  </style>
</head>
<body>
  <h1>Hello, world!</h1>
  <p>This is a paragraph before the div.</p>
  <div class="mydiv">
    <p>This is a paragraph inside the div.</p>
  </div>
  <p>This is a paragraph after the div.</p>
  <div class="mydiv">
    <p>This is a paragraph inside the second div.</p>
  </div>
  <script>
    console.log(" Hello from JS!");
    document.getElementsByClassName("mydiv")[0].style.backgroundColor = "green";
  </script>
</body>
</html>
```

JavaScript in HTML

- JavaScript controls behavior
- Add with `<script>` tag

Hello, world!

This is a paragraph before the div.

This is a paragraph inside the div.

This is a paragraph after the div.

This is a paragraph inside the second div.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hello</title>
  <style>
    .mydiv { color: orange; }
  </style>
</head>
<body>
  <h1>Hello, world!</h1>
  <p>This is a paragraph before the div.</p>
  <div class="mydiv">
    <p>This is a paragraph inside the div.</p>
  </div>
  <p>This is a paragraph after the div.</p>
  <div class="mydiv">
    <p>This is a paragraph inside the second div.</p>
  </div>
  <script>
    console.log(" Hello from JS!");
    let divs = document.getElementsByClassName("mydiv");
    for (let div of divs) {
      div.style.backgroundColor = "green";
    }
  </script>
</body>
</html>
```

JavaScript in HTML

- JavaScript controls behavior
- Add with `<script>` tag

Hello, world!

This is a paragraph before the div.

This is a paragraph inside the div.

This is a paragraph after the div.

This is a paragraph inside the second div.

Change Color

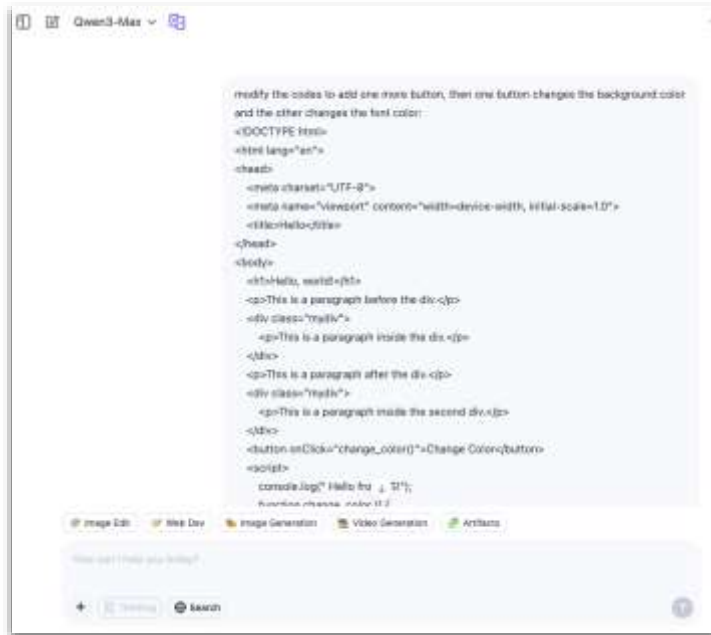
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hello</title>
</head>
<body>
  <h1>Hello, world!</h1>
  <p>This is a paragraph before the div.</p>
  <div class="mydiv">
    <p>This is a paragraph inside the div.</p>
  </div>
  <p>This is a paragraph after the div.</p>
  <div class="mydiv">
    <p>This is a paragraph inside the second div.</p>
  </div>
  <button onClick="change_color()">Change Color</button>
  <script>
    console.log(" Hello from JS!");
    function change_color () {
      let divs = document.getElementsByClassName("mydiv");
      for (let div of divs) {
        div.style.backgroundColor = ["green", "red", "yellow"][Math.floor(Math.random() * 3)];
        div.style.color = ["black", "blue", "white"][Math.floor(Math.random() * 3)];
      }
    }
  </script>
</body>
</html>
```

Add More Functions with AI

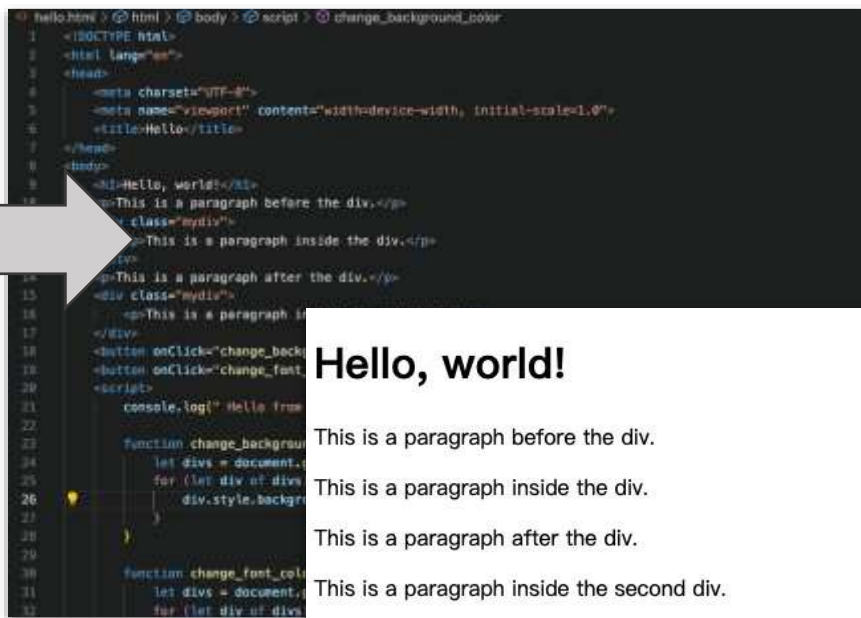
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hello</title>
</head>
<body>
  <h1>Hello, world!</h1>
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  </div>
  <p>This is a paragraph after the div.</p>
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    <p>This is a paragraph inside the second div.</p>
  </div>
  <button onClick="change_color()">Change Color</button>
</script>
  console.log(" Hello from JS!");
  function change_color () {
    let divs = document.getElementsByClassName("mydiv");
    for (let div of divs) {
      div.style.backgroundColor = ["green", "red", "yellow"][Math.floor(Math.random() * 3)];
      div.style.color = ["black", "blue", "white"][Math.floor(Math.random() * 3)];
    }
  }
</script>
</body>
</html>
```

modify the codes to add one more button, then one button changes the background color and the other changes the font color:

{the copied lines}



Add More Functions with AI



Hello, world!

This is a paragraph before the div.

This is a paragraph inside the div.

This is a paragraph after the div.

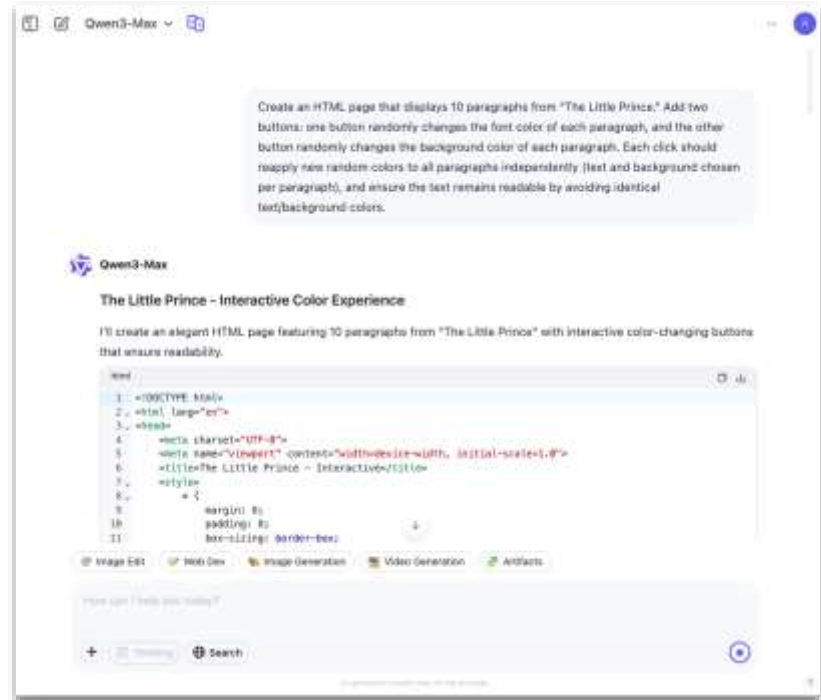
This is a paragraph inside the second div.

Change Background Color

Change Font Color

Write All Codes with AI

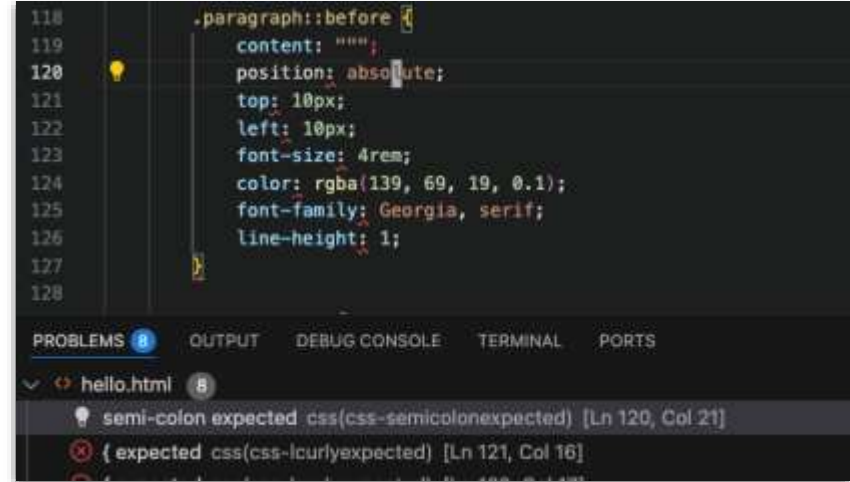
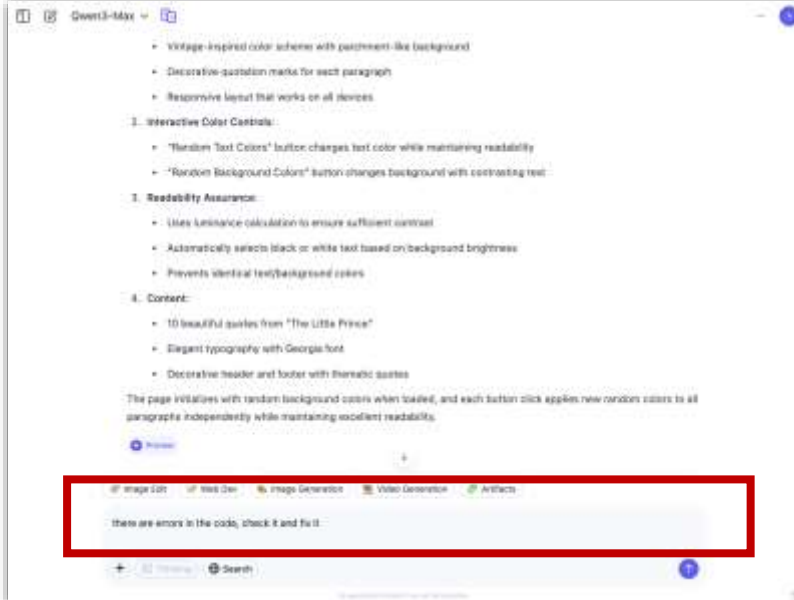
Create an HTML page that displays 10 paragraphs from “The Little Prince.” Add two buttons: one button randomly changes the font color of each paragraph, and the other button randomly changes the background color of each paragraph. Each click should reapply new random colors to all paragraphs independently (text and background chosen per paragraph), and ensure the text remains readable by avoiding identical text/background colors.



Fix Bugs with AI

Ask the AI to fix any errors in your content. The more information and context you provide, the more accurate and helpful its corrections will be.

There are errors in the code. Check it and fix it.



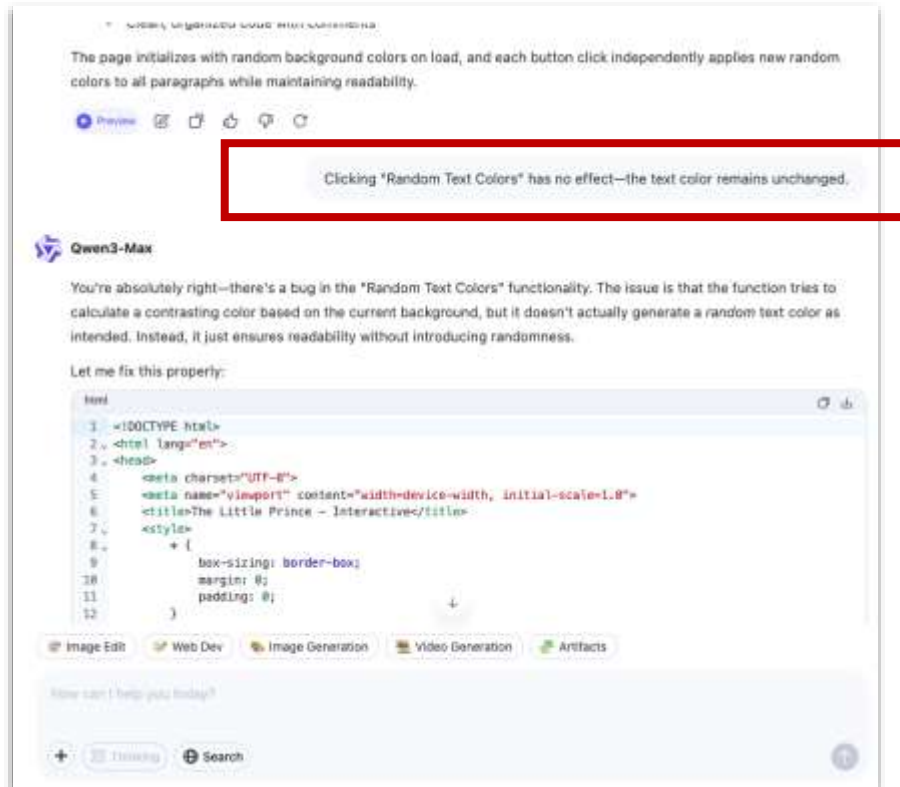
There are error in line 120: semi-colon expected, which is the third line of following code block:

```
.paragraph::before {
  content: "";
  position: absolute;
  top: 10px;
  left: 10px;
  font-size: 4rem;
  color: rgba(139, 69, 19, 0.1);
  font-family: Georgia, serif;
  line-height: 1;
}
```

Fix Bugs with AI

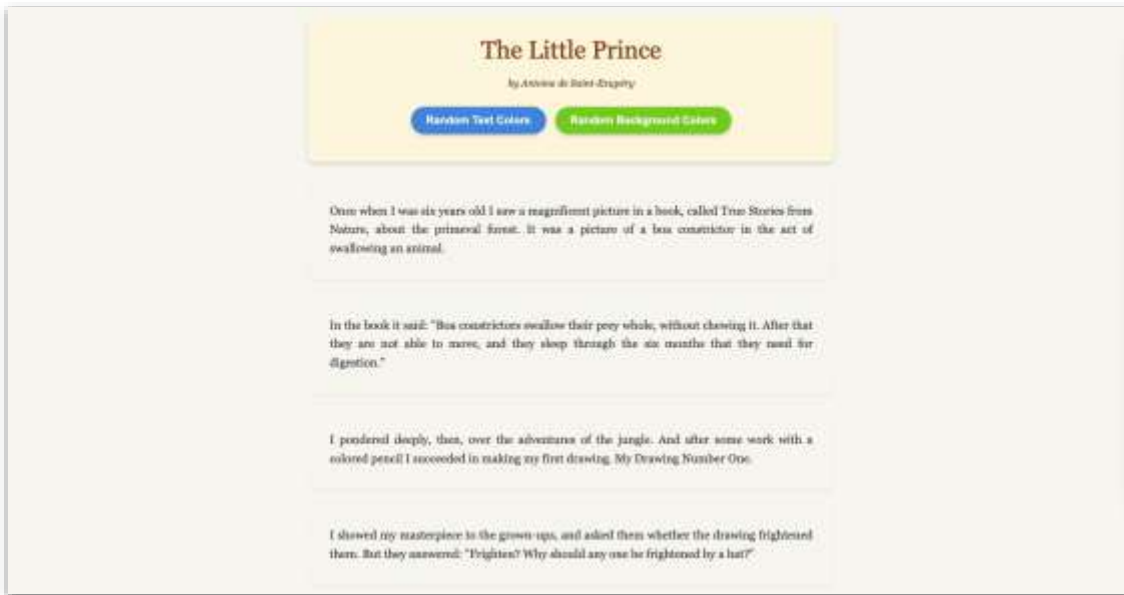


Clicking “Random Text Colors” has no effect—the text color remains unchanged.

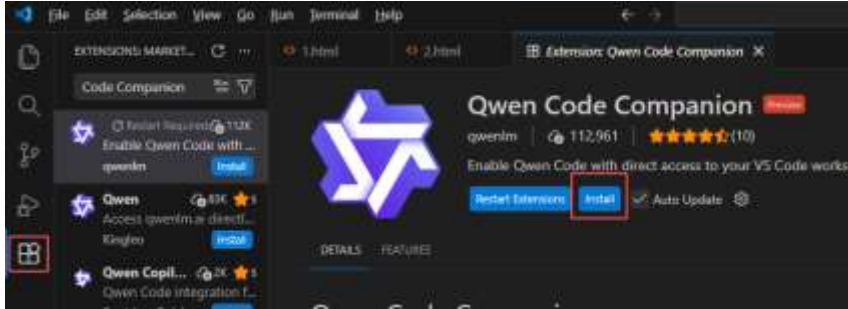


Summary

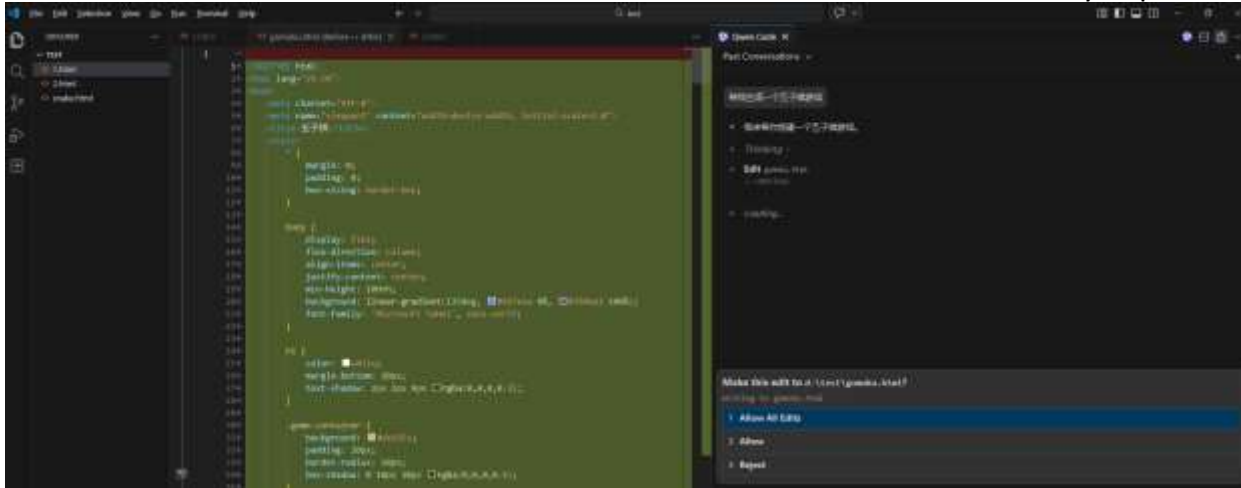
- Have a look at HTML, CSS, and Javascript.
- Write all codes for hello.html with AI.



Qwen Code Plugin



1. Open VSCODE and click on "Extensions" on the left side.
2. Enter "Qwen Code Companion" in the search box, click the first option, and then click the Install button.
3. After installation, on macOS, press Cmd + Shift + P (on Windows, press Ctrl + Shift + P) to open the command panel. Enter "Qwen Code: Open Chat" and press Enter to open the dialog box.
4. Navigate to your browser and confirm the use.
5. After generating the code, simply confirm and the code will be automatically copied



More Vibe Coding

Make the page looking more beautiful.



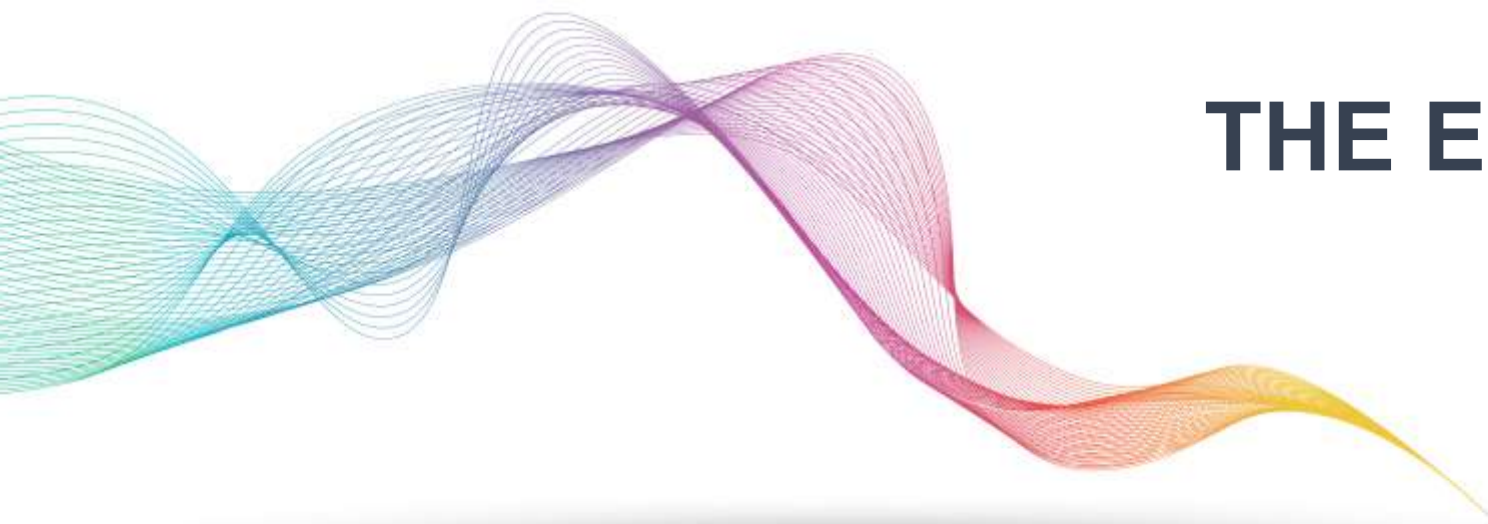
Add some animations to the page.



More Vibe Coding

Create an HTML page that displays **{add your idea}**

- Add buttons: **{describe what you want to do when clicking on the buttons}**
- Add animations: **{describe what animations you like}**
- **{other requirements}**



THE END